

Database management system

Q1. What is a data.?

Ans: A data is collection of raw material. That are store at the time of beginning that are may be useful or not be useful. For student information: std_name, class, subject, address, mobile_no

Q2. What is information.?

Ans: An information is summarized and resultant data that are must be useful.
For student marksheet: std_name, class, subject, marks

Q3. What is a File System. Explain its.?

Ans: A file system is collection of different data and information. It is a manual file system was used to maintain the records and files. Data were store and processed by using different method and procedures that are based upon traditional file system. The number of files exist in file system that are not connected to each other. It is known as file system.

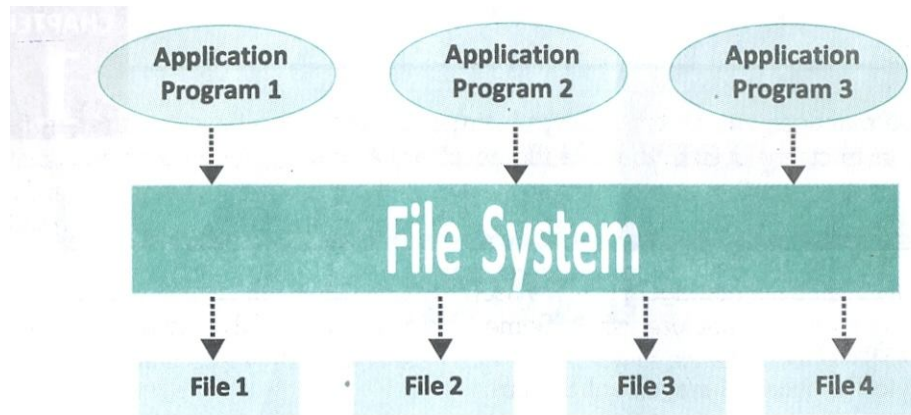


Fig: File System

Disadvantages of File System

1. **Data Inconsistency:** Same data occur in multiple locations.
2. **No Data Security:** There is no password facility in file system so that data is not more secure.
3. **No data integrity:** The number of files exist in file system that are not connected to each other.
4. **Data accessibility depends upon the person:** Data store is based upon manual file system so that data accessibility depends upon the person.
5. **No data backup and recovery:** data store manually in the file so that there is no data backup and recovery facility.

Q4. What is Database. Explain its.

Ans: A Database is a collection of meaningful data and information that are related to each others. Like that STUDENT, EMPLOYEE, TEACHER, STAFF

Q3. What is Database Management System. Explain its advantages and disadvantages.?

Ans: A DBMS is a collection of meaningful data and information that are related to each other that are manage by specific software and application program. Like that MYSQL, ORACLE, SYBASE, DB2.

Advantages of DBMS

1. **Data Sharing:** The most important feature of DBMS is data sharing. All the file and records are connected to each other. They are share the data each other.
2. **No Data Duplicacy:** In DBMS, Data is store with the help of key attributes so that there is no possibility data duplicacy.
3. **Data integrity:** In DBMS, Data is store within table so that they are connected to each other.
4. **High reliability:** In DBMS, A single copy of record exist in a multiple location. If there is some problem in a one location the data can be access from one other location.
5. **Data Consistency:** A correct and relevant data store in DBMS.
6. **Back and Recovery:** In DBMS, Data is store with the help of specific software and application program so that there is a feature available for data back and recovery.
7. **Easy to access:** In DBMS, Data is store in table with the help of key attribute. Data can be access easily with the help of SQL language.
8. **Security:** DBMS provides password facility so that data is more secure.

Disadvantage of DBMS:

1. Highly technical person is required for managing DBMS.
2. There are more possibility to attack different attacker and virus.
3. It is very difficult to manage large DBMS.

Q5. Write the difference between File Management System and DBMS.?

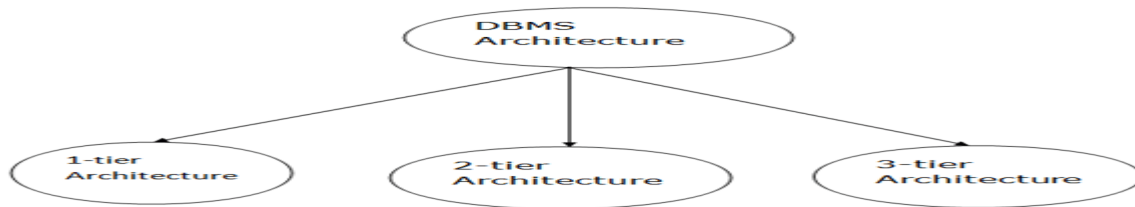
Ans:

S.No.	File Management System	Database Management System
1.	Data store and manage with the help of procedure and method.	Data store and manage with the help of specific software and application program.
2.	Data is store within file.	Data is store within table.
3.	More possibility to data duplicacy.	No possibility to data duplicacy.
4.	No data integrity.	Data integrated to each other.
5.	Less data security.	More data security.
6.	No data backup and recovery features.	Available data backup and recovery feature.

Q6. What is DBMS Architecture.?

- The DBMS design depends upon its architecture. DBMS architecture depends upon how users are connected to the database to get their request done.

Types of DBMS Architecture

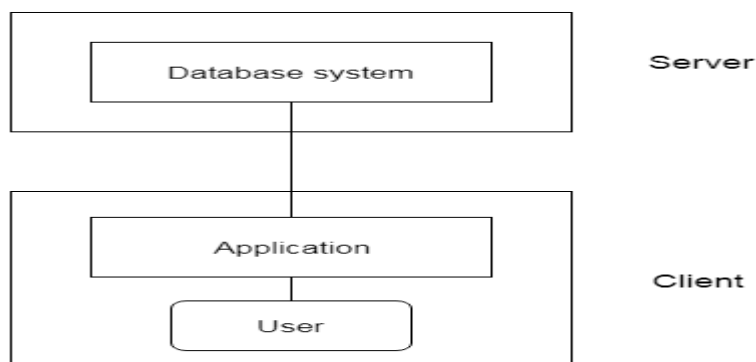


1-Tier Architecture

- In this architecture, the database is directly available to the user. It means the user can directly sit on the DBMS and uses it.
- Any changes done here will directly be done on the database itself. It doesn't provide a handy tool or interface for end users.
- The 1-Tier architecture is used for development of the local application, where programmers can directly communicate with the database for the quick response.

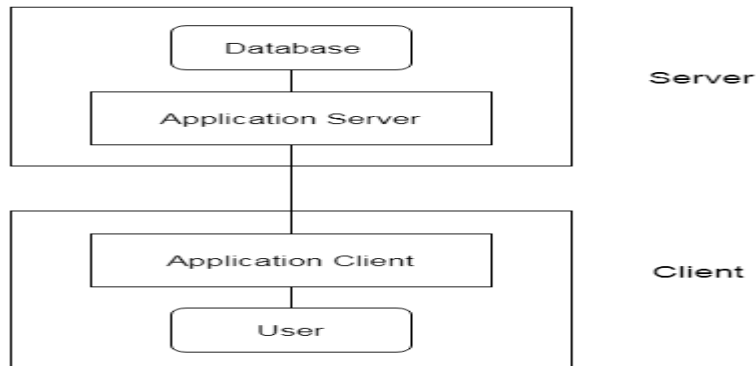
2-Tier Architecture

- The 2-Tier architecture is based client-server architecture. In the two-tier architecture, applications on the client end can directly communicate with the database at the server side. The user interfaces and application programs are run on the client-side.
- The server side is responsible to provide the functionalities like: query processing and transaction management.
- To communicate with the DBMS, client-side application establishes a connection with the server side.



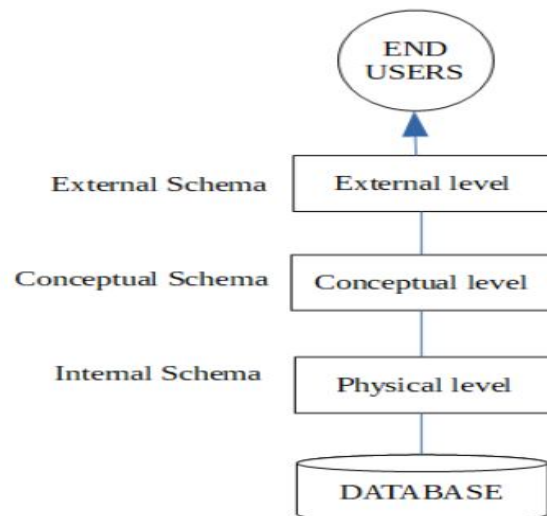
3-Tier Architecture

- The 3-Tier architecture contains another layer between the client and server. In this architecture, client can't directly communicate with the server.
- The application on the client-end interacts with an application server which further communicates with the database system.
- End user has no idea about the existence of the database beyond the application server. The database also has no idea about any other user beyond the application.
- The 3-Tier architecture is used in case of large web application.



Q7.What is Three level Architecture of DBMS.?

Ans: The three levels present in this architecture are Physical level, Conceptual level and External level.



The details of these levels are as follows –

Physical Level

This is the lowest level in the three level architecture. It is also known as the internal level. The physical level describes how data is actually stored in the database. In the lowest level, this data is stored in the external hard drives in the form of bits and at a little high level, it can be said that the data is stored in files and folders.

Conceptual Level

The conceptual level is at a higher level than the physical level. It is also known as the logical level. It describes how the database appears to the users conceptually and the relationships between various data tables. The conceptual level does not care for how the data in the database is actually stored.

External Level

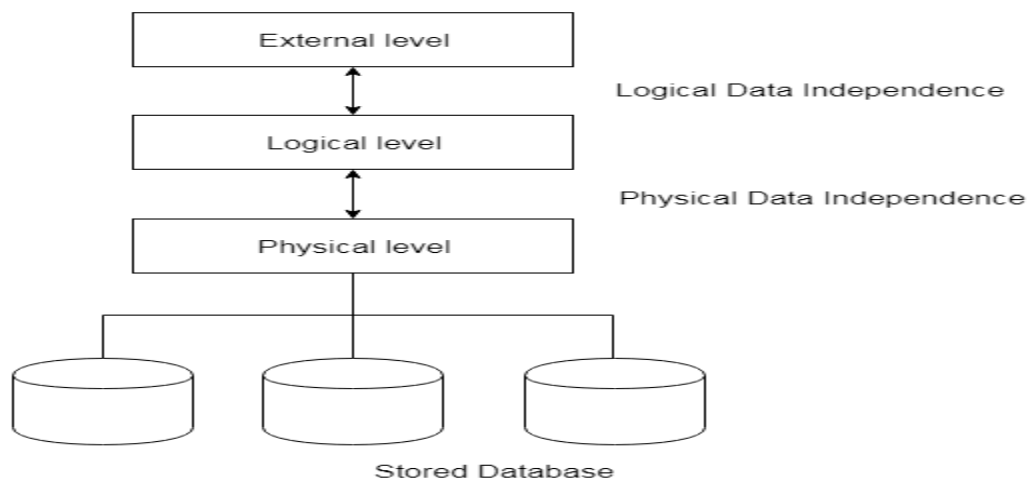
This is the highest level in the three level architecture and closest to the user. It is also known as the view level. The external level only shows the relevant database content to the users in the form of views and hides the rest of the data. So different users can see the database as a different view as per their individual requirements.

Q8. What is Data Independence.? Explain its

Ans: Data independence is the ability to modify the scheme without affecting the programs and the application to be rewritten. Data is separated from the programs, so that the changes made to the data will not affect the program execution and the application.

There are two levels of data independence based on three levels of abstraction. These are as follows –

- Physical Data Independence
- Logical Data Independence



Physical Data Independence

Physical Data Independence means changing the physical level without affecting the logical level or conceptual level.

It may include changes using the following –

- A new storage device like magnetic tape, hard disk, etc.
- A new data structure for storage.
- Changing the location of the database.
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Logical Data Independence

Data Independence says that users should be able to manipulate the Logical View of data without any information of its physical storage. Software or the computer program is used to manipulate the logical view of the data.

Database administrator is the one who decides what information is to be kept in the database and how to use the logical level of abstraction. It provides the global view of Data. It also describes what data is to be stored in the database along with the relationship.

Q9. What is Schemas and instances in DBMS.?

Ans: The overall design of the database is called database schema. It is the logical structure of a database. Schema will not be changed frequently. It does not show the data in the database.

Types of Schema

The different types of schemas are as follows –

- **Physical schema**
- **Logical schema**
- **External**

Example

Std_Name	Email	Phone_No
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Instance

Instance is a collection of information that stored in a database at a particular moment is called an instance of the database. It is a dynamic value which keeps on changing.

Example

Std_Name	Email	Phone_No
Amit	amit@gmail.com	982456232

Q10. What is a Data Dictionary.

Ans: Data Dictionary is a collection of names, definitions, and attributes. It is a meta data. It means that it is data about data that are related to particular field.

Std_Name	Email	Phone_No
Amit	amit@gmail.com	982456232

Q11. What is Database User.?

Ans: Database users are categorized based up on their interaction with the database. These are seven types of database users in DBMS.

1. **Database Administrator (DBA) :** Database Administrator (DBA) is a person/team who create data base and who has a full authorization of database. DBA will create a new account id and password for the user if he/she need to access the database. DBA is also responsible for providing security to the database and he allows only the authorized users to access/modify the data base. .
2. **Naive / Parametric End Users :** Parametric End Users are database user who don't have any DBMS knowledge but they frequently use the database applications in their daily life to get the desired results. For examples, Railway's ticket booking Agent
3. **System Analyst :**
System Analyst is a user who analyzes the requirements of parametric end users. They check whether all the requirements of end users are satisfied.
4. **Sophisticated Users :** Sophisticated users can be engineers, scientists, business analyst, who are familiar with the database. They can develop their own database applications according to their requirement.
5. **Database Designers :** Data Base Designers are the users who design the structure of database which includes tables and database structure.
6. **Application Programmers :** Application Programmers also referred as System Analysts or simply Software Engineers, are the back-end programmers who writes the code for the application programs. They are the computer professionals. These programs could be written in Programming languages such as Visual Basic, Developer, C, FORTRAN, COBOL etc
7. **Casual Users / Temporary Users :** Casual Users are the users who occasionally use/access the database but each time when they access the database they require the new information, for example, Middle or higher level manager.